

9	(a) $V_B = 1000 \text{ mV}$ when strained, $V_A = 2000 \times 121.5 / (121.5 + 120.0)$ $= 1006.2 \text{ mV}$ change = 6.2 mV (<i>allow 6 mV</i>)	C1	M1	A1	[3]
(b)	(i) 1. resistor between V_{IN} and V^- and V^+ connected to earth resistor between V^- and V_{OUT}	B1	B1	[2]	
	2. P / + sign shown on earth side of voltmeter	B1		[1]	
	(ii) ratio of $R_F / R_{IN} = 40$ R_{IN} between 100Ω and $10 \text{ k}\Omega$ (<i>any values must link to the correct resistors on the diagram</i>)	M1	A1	[2]	